

Practical Experiment 4

Question 1:

Code:

```
#include <stdio.h>

#include <unistd.h>

#include <sys/wait.h>


int main()

{

    char command[50];


    printf("Enter command: ");

    scanf("%s", command);


    int pid = fork();


    if (pid == 0)

    {

        printf("Child PID: %d\n", getpid());

        execlp(command, command, NULL);

    }

    else

    {
```

```

printf("Parent PID: %d\n", getpid());

printf("Child PID: %d\n", pid);

wait(NULL);

printf("Child completed\n");

}

return 0;

}

```

Output :

```

/Users/rishivarma/Programs/Practical\ Exp\ 4/exp1 / exit;
rishivarma@rishis-MacBook-Air ~ % /Users/rishivarma/Library/Mobile Documents/com~apple~/CloudDocs/Documents/Documents
\ -\ rishi's MacBook Air/College/OSSP/Programs/Practical\ Exp\ 4/exp1 ; exit;
Enter command: ls
Parent PID: 6509
Child PID: 6510
Child PID: 6510
a.out          hello.c        Pictures       Untitled.ipynb
Applications   hello2.c       Public        Untitled1.ipynb
Book 1(Sheet1).csv housing.csv    PyCharmMiscProject  Untitled2.ipynb
copy.txt       input.txt     read          Untitled3.ipynb
Desktop        Library      read.c       Untitled4.ipynb
Documents      ml-env       sklearn-env
Downloads      Movies
folder         Music
Child completed

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]

```

Question 2:

Code:

```

#include <stdio.h>

#include <unistd.h>

#include <sys/wait.h>

int main()

```

```
{  
  
    char command[50];  
  
    printf("Enter command: ");  
    scanf("%s", command);  
  
    int pid = fork();  
  
    if (pid == 0)  
    {  
        printf("Child PID: %d\n", getpid());  
        execlp(command, command, NULL);  
    }  
    else  
    {  
        printf("Parent PID: %d\n", getpid());  
        printf("Child PID: %d\n", pid);  
  
        wait(NULL);  
  
        printf("Child completed\n");  
    }  
  
    return 0;  
}
```

Output:

```
/Users/rishivarma/Library/Mobile Documents/com~apple~CloudDocs/Documents/Documents
rishivarma@rishis-MacBook-Air ~ % /Users/rishivarma/Library/Mobile Documents/com~apple~CloudDocs/Documents/Documents
-\ rishi's MacBook Air/College/OSSP/Programs/Practical\ Exp\ 4/exp2 ; exit;
File copied successfully

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

Question 3:

Code:

```
#include <stdio.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
int main()
```

```
{
```

```
    int source, destination;
```

```
    char buffer[100];
```

```
    int n;
```

```
    source = open("source.txt", O_RDONLY);
```

```
    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
```

```
    n = read(source, buffer, sizeof(buffer));
```

```
write(destination, buffer, n);

close(source);

close(destination);

printf("File copied successfully\n");

return 0;

}
```

Output:

```
rishivarma@rishis-MacBook-Air ~ % /Users/rishivarma/Library/Mobile Documents/com~apple~CloudDocs/Documents/Documents
\ -\ rishi's\ MacBook\ Air\College/OSSP/Programs/Practical\ Exp\ 4/exp3 ; exit;
Parent Process
PID: 6561
Child PID: 6562
Child Process
PID: 6562
PPID: 6561
Child completed
Parent completed

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

Question 4:

Code:

```
#include <stdio.h>

#include <unistd.h>

#include <sys/wait.h>
```

```
int main()

{

    int pid = fork();


    if (pid == 0)
    {

        printf("Child Process\n");

        printf("PID: %d\n", getpid());

        printf("PPID: %d\n", getppid());


        sleep(5);


        printf("Child completed\n");
    }

    else

    {

        printf("Parent Process\n");

        printf("PID: %d\n", getpid());

        printf("Child PID: %d\n", pid);


        wait(NULL);


        printf("Parent completed\n");
    }


    return 0;
```

```
}
```

Output:

```
rishivarma@rishis-MacBook-Air ~ % /Users/rishivarma/Library/Mobile\ Documents/com~apple~CloudDocs/Documents/Documents
\ -\ rishi's\ MacBook\ Air/College/OSSP/Programs/Practical\ Exp\ 4/exp4 ; exit;
Child 1 PID: 6587
Parent PID: 6586
Child 2 PID: 6588
Child 1 completed
One child completed
Child 2 completed
Second child completed

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```